THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 5: **Exploring Forces** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A force is a: 1

(a) type of energy only **(b)** push or a pull
(c) kind of speed **(d)** state of rest

2. Friction is greater between: 1

(a) smooth surfaces **(b)** polished surfaces
(c) oiled surfaces **(d)** rough surfaces

3. A body is charged by rubbing. The two types of charges are: 1

(a) heavy and light **(b)** positive and negative
(c) hot and cold **(d)** north and south

4. The SI unit of force is the: 1

(a) pascal **(b)** joule **(c)** newton **(d)** watt

5. Which of these reduces friction? 1

(a) rough paper **(b)** dust **(c)** sand **(d)** lubricants

6. Like charges: 1

(a) become neutral **(b)** attract each other
(c) do not interact **(d)** repel each other

7. A force can change: 1

(a) only the temperature
(b) the speed, direction or shape of an object
(c) only the colour
(d) only the mass

8. Compared with sliding friction, rolling friction is: 1

(a) the same **(b)** twice as much **(c)** more **(d)** less

9. Unlike charges: 1

(a) repel each other **(b)** attract each other
(c) do not interact **(d)** vanish

10. Which of these is a contact force? 1

(a) electrostatic force **(b)** gravitational force
(c) friction **(d)** magnetic force

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SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. What is a force? State its SI unit. 2

12. Why do we sprinkle powder on a carrom board? 2

13. State the law of charges. 2

14. Differentiate between contact and non-contact forces with one example each. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how force can change the shape of an object. Give two examples. 3

16. Describe how an electroscope works. 3

17. A force of 5 N and another of 5 N act on a book in opposite directions. What is the net force and the effect? 3

OR

Describe the types of friction with an example for each. 3

18. Explain how a body can get charged by rubbing. Give an example. 3

OR

Describe an activity to show that magnets attract some materials and not others. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the effects of force with five examples. 5

OR

Explain the working of a gold-leaf electroscope and what happens when a charged rod touches or is brought near the disc. 5

20. Explain the types of forces: contact and non-contact, with at least two examples for each. 5

OR

Explain the gravitational force on Earth. How does it differ on the Moon? Discuss mass and weight. 5

END OF QUESTION PAPER — Total: 40 Marks

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